

CBE 30235: Introduction to Nuclear Engineering

Problem Set 4: SOLUTIONS

Interactions, Moderation, and Fission

Problem 1: Shielding Design (^{10}B)

Given:

- $I_0 = 10^{10}$ n/cm²s, $I_{\text{target}} = 10^3$ n/cm²s.
- Material: Pure ^{10}B . $\sigma_a = 3840$ b. $\rho = 2.34$ g/cm³.
- Atomic Mass of $^{10}\text{B} \approx 10.0$ g/mol.

(a) Calculate the atom density (N)

$$N = \frac{\rho N_A}{M} = \frac{(2.34 \text{ g/cm}^3)(6.022 \times 10^{23} \text{ atoms/mol})}{10.0 \text{ g/mol}}$$

$$\mathbf{N = 1.41 \times 10^{23} \text{ nuclei/cm}^3}$$

(b) Calculate Macroscopic Cross Section (Σ_a)

$$\Sigma_a = N\sigma_a = (1.41 \times 10^{23} \text{ cm}^{-3})(3840 \times 10^{-24} \text{ cm}^2)$$

$$\mathbf{\Sigma_a = 541.4 \text{ cm}^{-1}}$$

(c) Mean Free Path (λ)

$$\lambda = \frac{1}{\Sigma_a} = \frac{1}{541.4 \text{ cm}^{-1}}$$

$$\lambda = \mathbf{0.00185 \text{ cm}} \quad (\approx \mathbf{18.5 \mu\text{m}})$$

Note: This illustrates why Boron is such an incredible absorber. Neutrons barely penetrate the surface.

(d) Required Thickness (x) We use the attenuation law $I(x) = I_0 e^{-\Sigma_a x}$.

$$\frac{I(x)}{I_0} = \frac{10^3}{10^{10}} = 10^{-7}$$

$$e^{-\Sigma_a x} = 10^{-7} \implies -\Sigma_a x = \ln(10^{-7}) = -16.12$$

$$x = \frac{16.12}{541.4 \text{ cm}^{-1}}$$

$$\mathbf{x = 0.0298 \text{ cm}} \quad (\approx \mathbf{0.3 \text{ mm}})$$

Problem 2: Moderation Mechanics

Given: $E_0 = 2 \text{ MeV}$ ($2 \times 10^6 \text{ eV}$), $E_{th} = 0.025 \text{ eV}$.

(a) **Collision Parameter α** Formula: $\alpha = \left(\frac{A-1}{A+1}\right)^2$

- **Hydrogen** ($A = 1$): $\alpha = ((1 - 1)/(1 + 1))^2 = 0$
- **Carbon** ($A = 12$): $\alpha = (11/13)^2 = 0.716$
- **Iron** ($A = 56$): $\alpha = (55/57)^2 = 0.931$

(b) **Minimum Energy after 1 collision with Carbon** The maximum energy loss corresponds to scattering to αE_0 .

$$E_{min} = \alpha_C E_0 = (0.716)(2 \text{ MeV})$$

$$E_{min} = 1.432 \text{ MeV}$$

(c) **Average Logarithmic Decrement ξ** Formula: $\xi \approx \frac{2}{A+2/3}$ (Exact $\xi = 1$ for H).

- **H:** $\xi = 1.0$
- **C:** $\xi \approx 2/(12.667) = 0.158$
- **Fe:** $\xi \approx 2/(56.667) = 0.035$

(d) **Average Number of Collisions (n)** Total lethargy change required: $\Delta u = \ln(E_0/E_{th}) = \ln(2 \times 10^6/0.025) = \ln(8 \times 10^7) = 18.2$.

$$n = \frac{\Delta u}{\xi}$$

- **H:** $18.2/1.0 \approx 18$ collisions
- **C:** $18.2/0.158 \approx 115$ collisions
- **Fe:** $18.2/0.035 \approx 520$ collisions

Conclusion: Iron is a poor moderator because its high mass ($A = 56$) means $\alpha \approx 1$. The neutron bounces off like a ping-pong ball hitting a bowling ball, losing very little energy per hit. It would take over 500 collisions to thermalize, during which time the neutron is likely to be absorbed (parasitic capture).

Problem 3: Reactor Mass Balance

Given: $P_{th} = 3000 \text{ MW}$. $E_R = 200 \text{ MeV/fission}$. Enrichment = 4.5 w/o ^{235}U .

(a) **Fission Rate** Convert units to Joules:

$$P = 3000 \text{ MW} = 3.0 \times 10^9 \text{ J/s}$$

$$E_R = 200 \text{ MeV} \times (1.602 \times 10^{-13} \text{ J/MeV}) = 3.204 \times 10^{-11} \text{ J/fission}$$

$$\text{Rate} = \frac{P}{E_R} = \frac{3.0 \times 10^9}{3.204 \times 10^{-11}}$$

$$\text{Rate} = 9.36 \times 10^{19} \text{ fissions/s}$$

(b) ^{235}U Consumption Rate (g/day)

$$\text{Atoms/day} = (9.36 \times 10^{19} \text{ s}^{-1}) \times (86400 \text{ s/day}) = 8.09 \times 10^{24} \text{ atoms/day}$$

$$\text{Mass/day} = \frac{8.09 \times 10^{24}}{N_A} \times 235 \text{ g/mol} = \frac{8.09 \times 10^{24}}{6.022 \times 10^{23}} \times 235$$

$$\text{Mass burned} = \mathbf{3,157 \text{ g/day} \approx 3.16 \text{ kg/day}}$$

(c) UO_2 Fuel Usage (kg/year) First, calculate total ^{235}U burned in one year:

$$M_{235, \text{year}} = 3.157 \text{ kg/day} \times 365 \text{ days} = 1,152 \text{ kg } ^{235}\text{U}$$

This ^{235}U makes up 4.5% of the Uranium mass in the fuel.

$$\text{Mass Uranium} = \frac{1,152 \text{ kg}}{0.045} = 25,600 \text{ kg U}$$

Now convert Mass U to Mass UO_2 . Assume $M_U \approx 238 \text{ g/mol}$ (approx) and $M_{\text{UO}_2} = 238 + 2(16) = 270 \text{ g/mol}$.

$$\text{Mass } \text{UO}_2 = 25,600 \text{ kg U} \times \frac{270}{238}$$

$$\text{Mass } \text{UO}_2 \approx \mathbf{29,040 \text{ kg/year} \approx 29 \text{ metric tons}}$$

(Note: This assumes we are calculating the amount of fuel fully depleted of its fissile content, which is a simplification. Real reactors only burn a fraction of the fuel before it must be replaced, but this mass balance is valid for the atoms actually fissioned).

Problem 4: Conceptual Physics

(a) Fissile vs. Fissionable

- ^{235}U is **Fissile**: The Binding Energy released by capturing a neutron ($\sim 6.5 \text{ MeV}$) is **greater** than the Critical Energy required to deform the nucleus. Therefore, even a zero-energy (thermal) neutron triggers fission.
- ^{238}U is **Fissionable**: The Binding Energy ($\sim 4.8 \text{ MeV}$) is **less** than the Critical Energy ($\sim 6 \text{ MeV}$). The neutron must bring the difference in the form of Kinetic Energy ($> 1 \text{ MeV}$) to cause the split.

(b) **Radioactivity of Fragments** Fission fragments are radioactive because they are **Neutron Rich**. The parent nucleus (^{235}U) has a high N/Z ratio (~ 1.55) to overcome proton repulsion. When it splits, the light fragments inherit this high ratio, but stable elements at $A = 95$ prefer a ratio of ~ 1.3 . They must undergo β^- decay (converting neutrons to protons) to reach stability.

(c) **Heat Deposition** The heat is generated **inside the fuel pellet**. This is because the heavy, highly charged fission fragments carry $\sim 80\%$ of the energy (168 MeV) but have a very short range ($\sim 10 \mu\text{m}$) due to strong Coulomb interactions with the fuel lattice. They stop and deposit their energy as heat before they can exit the fuel grain.

Problem 5: Enrichment and η

Formula: $\eta = \nu \frac{\Sigma_f}{\Sigma_a} = \nu \frac{N_{235}\sigma_{f,235} + N_{238}\sigma_{f,238}}{N_{235}\sigma_{a,235} + N_{238}\sigma_{a,238}}$ Since $N \propto$ atom fraction (x), and $\sigma_{f,238} = 0$, this simplifies to:

$$\eta = \nu \frac{x_{235}\sigma_{f,235}}{x_{235}(\sigma_{f,235} + \sigma_{\gamma,235}) + x_{238}\sigma_{\gamma,238}}$$

Data: $\sigma_{a,235} = 582 + 99 = 681$ b. $\sigma_{a,238} = 2.7$ b.

(a) **Pure ^{235}U** ($x_{235} = 1, x_{238} = 0$)

$$\eta = 2.42 \times \frac{582}{681} = \mathbf{2.07}$$

(b) **Natural Uranium** ($x_{235} = 0.0072, x_{238} = 0.9928$)

$$\Sigma_f \propto 0.0072(582) = 4.19$$

$$\Sigma_a \propto 0.0072(681) + 0.9928(2.7) = 4.90 + 2.68 = 7.58$$

$$\eta = 2.42 \times \frac{4.19}{7.58} = \mathbf{1.34}$$

(c) **LEU** ($x_{235} = 0.04, x_{238} = 0.96$)

$$\Sigma_f \propto 0.04(582) = 23.28$$

$$\Sigma_a \propto 0.04(681) + 0.96(2.7) = 27.24 + 2.59 = 29.83$$

$$\eta = 2.42 \times \frac{23.28}{29.83} = \mathbf{1.89}$$

(d) **Feasibility** For Natural Uranium, $\eta = 1.34$. This is much lower than the required $\eta > 1.8$ for Light Water Reactors. In an LWR, ordinary water absorbs many neutrons (σ_a for H is 0.33 b). If your source term (η) is only 1.34, you cannot afford the losses to the water, the structural steel, and leakage. You would fall below criticality ($k < 1$). Enriching to 4% boosts η to 1.89, providing enough "excess neutrons" to afford the luxury of using water as a moderator/coolant.